

### I. FULL STRUCTURE TO DEVELOP

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#### ◆ A. Vision & Positioning

##### 1.1 Project Objective

- Provide a hybrid platform combining Preply (flexible private tutoring) and Moodle (structured pedagogical power).
- Target students, parents, teachers, schools, and companies.
- Integrate AI-based pedagogy and academic traceability.

##### 1.2 Business Model

- Commission on private lessons.
- Premium subscriptions (structured access + AI features).
- Paid AI services: automated evaluation, progress tracking, preparation tools.
- Group classes (priced per seat or package).
- B2B training offers (individual or collective).

##### 1.3 Key Differentiators

- Modular structure: free, semi-structured, and fully structured modes.
- Alignment with national academic programs.
- Smart tracking, AI evaluation, dashboards for parents/teachers.

##### 1.4 Core Offering

- Personalized academic support + AI-powered tools.
- Voice assistance + translation + full accessibility.

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#### ◆ B. Pedagogical Architecture (LMS)

##### 2.1 Three Learning Modes

- **Free Mode:** flexible courses, mutual agreement (Preply-style).
- **Semi-Structured Mode:** pre-course checklist with chapters to be ticked.
- **Structured Mode:** course template with content + IA assessment.

##### 2.2 Academic Path Models

- Per country: C7/C8 (Morocco), Grade 8 (Pakistan), 4e/3e (France), etc.
- Organized by year, semester, subject, chapter.

##### 2.3 Standard Course Template

- Course sheet.
- Corrected exercises.
- Pre-class quiz.
- Post-class quiz + IA revision sheet.

##### 2.4 Lesson Plan / Checklist

- Shared with student/parent.
- Feeds AI evaluation system (before/after).

##### 2.5 AI Evaluation Engine

- Generates: quizzes, summaries, flashcards, self-evaluations.
- Real-time correction, adaptive learning suggestions.

##### 2.6 Progress Tracking

- Dashboards: student, teacher, parent, admin.

- Reports per session / week / semester / year.

## **2.7 Virtual Live Class**

- WebRTC HD, whiteboard, transcription, AI summary.

## **2.8 Exam Preparation / Certification**

- AI-generated mock exams.
- Auto-correction + AI recommendations.

## **2.9 Group Classes**

- For schools or companies.
- "Classroom view" and centralized management.

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# ◆ **C. Technical Modules**

## **3.1 Frontend**

- Responsive (desktop/mobile/tablet).
- PWA / Offline-ready / Speed / Accessibility.

## **3.2 Backend**

- TypeScript stack (Next.js + Node.js).
- REST API + scalable DB (PostgreSQL).

## **3.3 Integrated AI**

- Student-teacher matching.
- Auto-generation: quizzes, summaries, flashcards.
- Teacher AI: content suggestions, quality alerts.

## **3.4 Virtual Class (LiveKit)**

- HD video, collaborative board, transcription, AI avatars.

## **3.5 Smart Scheduling**

- Sync with Google/iCal, time zone automation, smart suggestions.

## **3.6 Payment**

- Stripe / PayPal.
- Sessions, subscriptions, commissions.

## **3.7 Messaging**

- WebSocket, push notifications, AI moderation, voice synthesis.

## **3.8 Multilingual & Accessibility**

- FR, EN, AR, IT, ES... WCAG 2.1 AA compliance.
- Instant translation, localized formats, voice navigation.

## **3.9 Support System**

- Chat, ticketing, self-help documentation.

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# ◆ **D. User Management**

## **4.1 Account Creation**

- Student / Parent / Teacher / Admin / Company.

## **4.2 Teacher Verification**

- Profile, diplomas, intro video, interview.

## **4.3 Levels & Subjects Assignment**

- Filter by country, course type, validated level.

## **4.4 Role-Based Dashboards**

- Parent, student, teacher, school.

## **4.5 Access & Logs**

- Audit logs, granular permissions.

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## ◆ E. Admin Tools

### 5.1 Central Admin Dashboard

- Users, income, performance stats.

### 5.2 Teacher Quality Control

- Ratings, alerts, AI moderation.

### 5.3 AI Service Activation

- Module activation per subscription tier.

### 5.4 RGPD Data Export / Reports

### 5.5 Group Course & B2B Session Control

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## ◆ F. Compliance & Security

### 6.1 Regulations

- GDPR / COPPA / Digital Services Act.

### 6.2 Authentication

- 2FA, Google OAuth.

### 6.3 Encryption

- TLS 1.3, AES-256.

### 6.4 Backups + Logs

- Encrypted backups, complete logging.

### 6.5 ISO Preparation

- ISO/IEC 27001 readiness.

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## ◆ G. Educational Tools & Content

### 7.1 AI Generators

- Quizzes, flashcards, notes.

### 7.2 Shared Library

- Level-based resources.

### 7.3 Pre-Filled Blocks

- Help teachers create content faster.

### 7.4 AI Remediation

- Support tools for struggling learners.

### 7.5 External Assessment Tools

- Exam simulation, auto-correction, AI feedback.

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## ◆ H. Monetization & Premium Offers

### 8.1 Session Commissions

### 8.2 "Learn with AI" Module

### 8.3 Subscriptions (families / schools / companies)

### 8.4 Official Certifications

### 8.5 Group Course Packages

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## ◆ I. Communication & Marketing

### 9.1 SEO Multilingual Pages

### 9.2 Acquisition Funnels

### 9.3 Guided Onboarding

### 9.4 Social Media / Ambassadors

## 9.5 Promotional Materials

### ◆ J. B2B Strategy & Ecosystem

#### 10.1 Company Training Offers

#### 10.2 Sponsored Classes / School Packs

#### 10.3 Institutional Partnerships

#### 10.4 White-Label Usage

#### 10.5 API Integrations (Moodle, Google Ed)

## II. PLAN DETAIL

### A. Vision & Positioning

#### 1. Project Objective

DESTAVIA is a hybrid online educational platform combining the academic structure and automation potential of Moodle with the flexibility and human interaction of marketplace models like Preply. It targets:

- Students (from primary school to university)
- Parents seeking measurable results
- Teachers who want to teach in a professional framework
- Companies looking to train their employees (B2B)

The project aims to:

- Provide structured academic support aligned with official programs by country
- Offer independent AI-driven assessment (before and after each session)
- Ensure traceability of learning progress and teaching quality
- Allow pedagogical freedom (structured or custom courses)
- Integrate advanced automation, accessibility, and collaboration tools

#### 2. Business Model

DESTAVIA combines multiple revenue streams:

- Commission on private lessons (Preply-like)
- Premium subscriptions (access to structured courses + AI tools)
- Sales of automated evaluation packs
- Group classes (seat- or class-based pricing)
- B2B training packages (individual or group)
- Add-on services: certificates, progression reports, personalized AI

#### 3. Strategic Differentiators

| Feature               | Preply           | Moodle              | DESTAVIA                                          |
|-----------------------|------------------|---------------------|---------------------------------------------------|
| Pedagogical structure | None (free-form) | Strong but complex  | Mixed: structured / semi-structured / free        |
| AI assessments        | No               | Partial             | Full before/after with learning tracking & alerts |
| Academic alignment    | No               | Yes (but technical) | Yes – per country, level, semester                |

| Feature                 | Preply          | Moodle  | DESTAVIA                                         |
|-------------------------|-----------------|---------|--------------------------------------------------|
| Course plan interface   | No              | No      | Pre-filled checklist before session              |
| Business model          | Commission only | None    | Mixed: commissions + subscriptions + IA services |
| Teacher content freedom | Yes             | No      | Yes, within structured template                  |
| Progress dashboards     | No              | Limited | Student / teacher / parent / admin               |
| Group courses           | No              | Yes     | Yes                                              |
| B2B offers              | No              | Limited | Yes – individual and group                       |

#### 4. Core Offer

- One-on-one courses with pre-session plan
- Full academic paths (semester/year/official curricula)
- Choice between 3 modes: free-form / semi-structured (checklist) / structured (model)
- AI-powered assessments: pre-quiz / post-quiz / summary / flashcards
- Dashboards for all roles: student, teacher, parent, company
- Booking system + messaging + live classroom tools
- AI tools for progress tracking, recommendations, moderation, voice synthesis

#### 5. Pedagogical Modes – Student & Teacher Logic

##### Student Side:

- Free-form: course and content agreed with teacher (like Preply) – standard pricing
- Semi-structured: student sees checklist of sections to be taught – moderate pricing
- Structured: full pedagogical content provided – premium pricing

##### Teacher Side:

- Free-form teaching: fast onboarding, basic credential check
- Semi-structured teaching: uses platform-provided checklist – mid-level validation
- Structured teaching: provides full pedagogical content according to standard template – high validation level

#### 6. Teacher Selection & Validation Process

To ensure teaching quality and learner confidence:

- Teacher profile must include CV, diploma scans, and demo lesson or interview
- Admin validation assigns teacher role: free / semi-structured / structured
- Continuous quality monitoring via student feedback, AI tracking, and moderation
- Teachers can be promoted across modes based on their performance and consistency



## B. Pedagogical Architecture (LMS)

### 1. The Three Learning Modes (for Students & Teachers)

**DESTAVIA offers three pedagogical formats** designed to accommodate different learner needs and teacher profiles, while enabling rich academic tracking and automation.

| Mode                    | Student Experience                                                                 | Teacher Requirements                                   | Pricing  | IA Tools Available                                  |
|-------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------|----------|-----------------------------------------------------|
| <b>Free-form</b>        | Student and teacher define course content freely (like Preply)                     | Basic profile, validated credentials                   | Standard | Post-session tracking only                          |
| <b>Semi-structured</b>  | Student chooses topics from a pre-defined checklist before each session            | Teacher checks off chapters to teach from academic map | Moderate | Pre/post quizzes, flashcards, recommendations       |
| <b>Fully Structured</b> | Teacher delivers a standardized course plan: full content, worksheets, evaluations | Teacher submits full session plans, validated by admin | Premium  | Full AI suite: quiz generation, scoring, dashboards |

✓ All modes benefit from automated **progress dashboards**, **quality tracking**, and optional AI support.

## 2. Academic Track Models (by Country & Curriculum)

DESTAVIA supports **structured alignment with national curricula**, enabling platform-wide consistency and measurable progress.

| Country              | Levels                                                          | Format                            |
|----------------------|-----------------------------------------------------------------|-----------------------------------|
| <b>Morocco</b>       | C7, C8, C9, lycée                                               | Chapters per subject per semester |
| <b>France</b>        | 6e, 5e, 4e, 3e, lycée                                           | National Education guidelines     |
| <b>Pakistan</b>      | G7, G8, G9, G10, O/A Levels                                     | Local Board syllabi               |
| <b>International</b> | IB / Cambridge / Custom Adult Tracks Language, business, skills |                                   |

Each academic level includes:

- Subjects with semester-based breakdown
- Chapter/Section structure
- Suggested weekly session plans

## 3. Standard Course Template (Structured Mode)

For teachers delivering structured courses, a **standardized template** ensures consistency and automation.

Each lesson includes:

1. **Course sheet** (key concepts, objectives, materials)
2. **Interactive exercises** (automatically corrected)
3. **AI-generated pre-quiz**
4. **Live session**
5. **AI-generated post-quiz**
6. **Flashcards & revision summary**

✓ Compatible with Moodle course format and auto-generators.

## 4. Lesson Plan Interface (Checklist for Semi-Structured Mode)

For semi-structured mode, DESTAVIA provides a **dynamic checklist** per subject.

Before each lesson:

- The **teacher selects** the chapters/sections they will cover
- The **student/parent receives** the plan in advance
- The **system prepares** pre-quiz and tailored revision path

🧠 This checklist is key to triggering the IA modules and tracking learning accurately.

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## 5. AI Evaluation Engine (All Modes)

DESTAVIA's IA engine supports:

- **Pre-session diagnostic quiz** (auto-generated from selected topics)
- **Post-session quiz** (adaptive, with difficulty levels)
- **Flashcard generator** (based on errors and key concepts)
- **Revision summaries** for students and parents
- **Remediation suggestions** for struggling learners

Bonus: **Teacher assistant IA** can suggest:

- Resources to upload
- New course ideas
- Alerts on struggling students

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## 6. Progress Dashboards

Each user has a tailored dashboard:

**Role**     **Features**

**Student** Progress chart, revision list, AI feedback, upcoming sessions

**Teacher** Course plans, session reports, quiz analytics

**Parent** Weekly report, IA summary, alerts, comparison with peers

**Admin** Overview of all actors, session stats, performance KPIs

These dashboards are powered by real-time data and AI-generated insights.

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## 7. Virtual Classroom & Collaboration Tools

Each live session includes:

- **HD video (WebRTC)** via LiveKit or similar
- **Collaborative whiteboard**
- **Automatic transcription**
- **Session recording**
- **AI-generated session summary**
- **Augmented reality & avatar preview (R&D)**

Accessible via browser and mobile apps.

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## 8. Certification & Exam Preparation

DESTAVIA supports students in **academic and international exam prep**:

- AI-generated **mock exams**
- Auto-corrected essays and problems
- **Score prediction engine**
- Certification modules (paid or bundled)

Targeted tests: Brevet, Bac, O-Levels, A-Levels, IB, TOEFL, DELE, etc.

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## 9. Group Classes (Academic or Corporate)

DESTAVIA allows:

- **School classes** (e.g. full C8 math group)
- **Corporate sessions** (e.g. English for HR teams)
- **Dashboards per class** (attendance, progress)
- **Shared resources & schedule management**

Class managers (school admin or company HR) get tailored tools.

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## C. Technical Modules (Infrastructure & Features)

This section outlines how DESTAVIA's pedagogical goals are **enabled by cutting-edge technologies**, offering a seamless, secure, and scalable experience for learners, teachers, and admins.

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### 1. Frontend Architecture (UX/UI & Accessibility)

| Pedagogical Need                                                                                                                                                                      | Technical Implementation                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Multi-device access (desktop, tablet, mobile)                                                                                                                                         | Responsive design with <b>Tailwind CSS &amp; Next.js</b>                             |
| Fast page loads                                                                                                                                                                       | <b>Progressive Web App (PWA)</b> capabilities, preload, image optimization           |
| Offline readiness                                                                                                                                                                     | PWA offline caching + local data sync                                                |
| Accessibility (vision, mobility, cognitive)                                                                                                                                           | WCAG 2.1 AA compliance, screen reader support, high contrast UI, keyboard navigation |
| International formats                                                                                                                                                                 | Auto-detection of date/number/language formats (i18n)                                |
|  <b>Result:</b> smooth, fast, accessible experience for all users, even on low-end mobile devices. |                                                                                      |

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### 2. Backend & Infrastructure

| Pedagogical Need                                                                                                                                            | Technical Implementation                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Real-time data (quizzes, dashboard, attendance)                                                                                                             | <b>Node.js + PostgreSQL + Redis</b>                                               |
| Scalable sessions, tracking, and chat                                                                                                                       | <b>Event-driven architecture, WebSocket</b> for real-time                         |
| AI tools (matching, quizzes, summaries)                                                                                                                     | <b>Python microservices with ML models (HuggingFace, OpenAI)</b>                  |
| Course & content management                                                                                                                                 | <b>Headless LMS architecture</b> using <b>custom APIs + Moodle/GraphQL bridge</b> |
| Reliability / performance                                                                                                                                   | <b>Vercel + Render</b> deployment + global CDN (Brotli compression, Workers Edge) |
| Developer-friendly                                                                                                                                          | Full-stack TypeScript, GitHub actions, CI/CD pipelines                            |
|  <b>Designed for rapid feature iteration and horizontal scalability.</b> |                                                                                   |

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### 3. AI-Powered Features

DESTAVIA integrates multiple AI engines to **automate, support, and enhance learning**.

| AI Tool                         | Purpose                                                    |
|---------------------------------|------------------------------------------------------------|
| Student–teacher matching        | NLP-based profile matching (goals, style, level, schedule) |
| Pre/post quiz generation        | Based on topic/level, auto-scored                          |
| Flashcards & summaries          | Post-session automatic generation                          |
| Plagiarism detection            | During written exercises or uploads                        |
| Content suggestion for teachers | Next-topic suggestions, course plan alerts                 |
| Performance alerts              | Drop in quiz scores, late homework, missing sessions       |
| Voice synthesis + translation   | Multilingual content, vocal accessibility                  |
| Chat moderation                 | Anti-toxicity, auto-flag, AI intervention                  |

✅ All AI features are modular and configurable per course/teacher/subscription level.

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### 4. Virtual Classroom (LiveKit WebRTC)

| Feature                 | Implementation                                |
|-------------------------|-----------------------------------------------|
| HD video & audio        | WebRTC via <b>LiveKit SDK</b>                 |
| Shared whiteboard       | Open-source canvas (Miro/Excalidraw clone)    |
| Auto-transcription      | <b>Whisper AI</b> or Google Speech-to-Text    |
| Session recording       | Stream to cloud with automatic indexing       |
| AI session summary      | NLP-based summary generated after class       |
| Augmented avatars (R&D) | AI-driven AR layer for immersive environments |

✅ Ready for one-on-one or group class formats, accessible in-browser or via mobile.

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### 5. Scheduling & Smart Booking

| User Experience           | Technical Feature                                  |
|---------------------------|----------------------------------------------------|
| Cross-timezone clarity    | Automatic timezone sync via user locale            |
| Calendar sync             | Integration with <b>Google Calendar, iCal</b>      |
| Reminders & alerts        | Push, SMS, Email notifications                     |
| Intelligent suggestions   | AI proposes optimal times based on past attendance |
| Double-booking prevention | Hard rules + visual warning system                 |
| Company/school scheduling | Group calendars + shared class views               |

✅ Teachers and students manage sessions effortlessly; schools and HR departments can organize multi-user sessions.

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### 6. Payments & Transactions

| Need                              | Integration                                  |
|-----------------------------------|----------------------------------------------|
| Secure one-time payments          | <b>Stripe Checkout / PayPal REST</b>         |
| Subscriptions (family, school)    | Stripe recurring billing                     |
| Custom B2B invoicing              | Admin-triggered invoice system               |
| Refunds, partial payments         | Stripe logic and admin rules                 |
| Profit sharing (commission model) | Automated split between platform and teacher |

- ✔ All transactions are tracked and logged for transparency and compliance.

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## 7. Messaging System

| Feature              | Technology                                 |
|----------------------|--------------------------------------------|
| Real-time messaging  | WebSocket + pub/sub engine                 |
| Push notifications   | Web + mobile (Firebase Cloud Messaging)    |
| Email & SMS fallback | Integrated via Twilio / Sendgrid           |
| AI moderation        | NLP toxicity filter + auto flag/report     |
| Voice messages & TTS | Speech-to-text and text-to-speech embedded |

- ✔ Supports formal lesson communication and casual student-teacher messaging, with built-in safety.

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## 8. Internationalization (i18n) & Localization

| Supported Feature    | Implementation                                 |
|----------------------|------------------------------------------------|
| Languages            | FR, EN, AR, ES, IT – fully translated strings  |
| Localized data       | Currencies, date formats, RTL support          |
| Multilingual content | Auto-translation of materials via DeepL/OpenAI |
| Speech & audio       | Voice synthesis (Google/ElevenLabs)            |
| Accessibility tools  | Subtitles, screen reader compatibility         |

- ✔ Expandable to any language with minimal dev time.

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## 9. Support & Documentation

| Role           | Tools                                            |
|----------------|--------------------------------------------------|
| Student/parent | Contextual help, chatbot (AI-powered), ticketing |
| Teacher        | FAQ, training videos, live support               |
| Admin          | Audit logs, usage reports, error trace tools     |
| Developer      | API docs, test environments, CLI toolkit         |

- ✔ All roles supported with smart assistance and full transparency.



## D. User Management & Permissions

DESTAVIA's platform architecture is designed for **multi-role usage**, including learners, parents, teachers, admins, and institutional partners. The system provides **fine-grained permissions**, **role-based dashboards**, and **automated validation flows** to ensure both educational and operational quality.

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### 1. Account Creation & Onboarding

| Role                              | Registration Requirements                   | Workflow                                                                              |
|-----------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------|
| Student                           | Email or mobile verification                | Simple sign-up + onboarding survey (level, goals, preferred language, preferred mode) |
| Parent                            | Linked to student account                   | Access to dashboard, progress, payment control                                        |
| Teacher                           | CV, diplomas, intro video, availability     | Multi-step validation: profile + document + training module + admin review            |
| Institution (School / Enterprise) | Company details, admin user, payment method | Enables multi-user dashboard and bulk enrollment                                      |
| Platform Admin                    | Internal only                               | Full control over system modules and data                                             |

✓ All user types access role-specific features with unique dashboards and permissions.

## 2. Teacher Profile Validation (Multi-Level)

DESTAVIA enforces **progressive teacher validation** depending on the selected teaching mode.

| Teaching Mode   | Validation Requirements                                | Access Level                                          |
|-----------------|--------------------------------------------------------|-------------------------------------------------------|
| Free-form       | Basic credentials + profile review                     | Can offer custom courses, flexible pricing            |
| Semi-structured | Must agree to checklist structure + one content review | Can use IA-based quizzes + session planning           |
| Structured      | Must submit full pedagogical content + model lesson    | Can teach in fully traceable, curriculum-aligned mode |

✓ Teachers can evolve from one mode to another after consistent quality reviews.

## 3. Role-Based Dashboards

Each user sees a **custom dashboard** adapted to their needs:

| Role       | Dashboard Features                                                               |
|------------|----------------------------------------------------------------------------------|
| Student    | Booked sessions, upcoming quizzes, progress graph, IA feedback, messaging        |
| Parent     | Child's timeline, evaluations, billing, communication with teacher               |
| Teacher    | Session planning, content submission, student performance alerts, income reports |
| Admin      | Global overview, user management, AI moderation, platform metrics                |
| Enterprise | Team-level tracking, employee feedback, invoice management                       |

✓ Dashboards are reactive, multilingual, and personalized.

## 4. User Permissions & Access Control

| Area            | Access Rules                                              |
|-----------------|-----------------------------------------------------------|
| Session content | Student, teacher, and parent (read-only)                  |
| Payment data    | Admin + parent (read/write), teacher (read earnings only) |
| AI evaluations  | Student (full), teacher (analytics), parent (summary)     |

| Area                                                                                                                                   | Access Rules                                           |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Course content (structured)                                                                                                            | Teacher edit, admin validate, student view             |
| Messaging                                                                                                                              | Role-based logic (e.g. student ↔ teacher, admin ↔ all) |
|  All actions are logged and auditable for compliance. |                                                        |

5. Data Privacy & Audit Logs

| Feature                                                                                                                                      | Technical Implementation                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Data access supervision                                                                                                                      | Role-based ACLs (Access Control Lists)                                                |
| Audit logs                                                                                                                                   | All sensitive actions timestamped and stored (modifications, deletions, file uploads) |
| Consent management                                                                                                                           | GDPR / COPPA-compliant checkboxes + parental consent                                  |
| User data export                                                                                                                             | One-click data export (ZIP + PDF) on request                                          |
| Suspicious behavior alerts                                                                                                                   | AI-driven monitoring and admin notifications                                          |
|  Ensures transparency, traceability, and user data control. |                                                                                       |

6. Teacher Evolution & Promotion

DESTAVIA uses an **internal scoring system** for teacher performance:

| Metric                                                                                                                                                                                                       | Weight                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Student feedback & ratings                                                                                                                                                                                   | ★★★★☆                  |
| Timeliness & reliability                                                                                                                                                                                     | ✓ Attendance rate      |
| Content quality                                                                                                                                                                                              | Reviewed by admin & AI |
| Engagement (sessions taught)                                                                                                                                                                                 | 📊 Over time            |
| Consistency of structure use                                                                                                                                                                                 | 📋 Checklist adherence  |
| Promotion triggers:                                                                                                                                                                                          |                        |
| <ul style="list-style-type: none"> <li>From free-form → semi-structured after X sessions + 4.5★+ avg</li> <li>From semi-structured → structured upon submission of full course path for 1 subject</li> </ul> |                        |
|  Motivates teachers to improve over time with clear progression path.                                                     |                        |

◆ E. Admin Dashboard & Quality Control

1. Central Admin Console

- Unified interface for **real-time monitoring** of:
  - User registrations (student / parent / teacher / company)
  - Course bookings & completions
  - Revenue by source (sessions, subscriptions, AI add-ons)
  - Feedback loops and alert systems
- AI-powered alerts for anomalies (e.g. no-shows, low ratings, inconsistent pricing)

## 2. Teacher Quality Monitoring

- AI + human hybrid scoring of:
  - Student feedback & lesson reviews
  - Lesson punctuality / attendance logs
  - Content quality (structure usage, adherence to checklists)
- Triggers for:
  - Auto-warning or flagging of low-performing teachers
  - Profile freezing or demotion to lower mode (structured → semi → free)

## 3. Activation / Control of Premium Services

- Admin can:
  - Enable/disable “Learn with AI” module per user or group
  - Monitor AI usage statistics (quizzes, flashcards, summaries used per user)
  - Set price per session or per add-on (pre/post tests, progress reports, AI flashcards)
  - Define school- or company-wide AI bundles

## 4. User Data & Reporting

- Exports in PDF/CSV (per student, per class, per teacher, per region)
- Filters by:
  - Course type (structured, semi, free)
  - Teacher rating
  - Completion rate
  - Session frequency and attendance
- Monthly reports auto-generated and emailed to admins

## 5. B2B & Collective Session Oversight

- Track ongoing group courses for institutions or companies
- Control class size, schedules, and assigned teachers
- Enable classroom-specific evaluation reports
- Adjust permissions for institutional coordinators (can view but not edit sessions)

## 6. AI Governance Layer

- All AI modules are auditable:
  - Quizzes have source logs (chapter ↔ question mapping)
  - Flashcards and summaries timestamped and stored
  - Admin review of AI-generated suggestions before publication (optional)
- Compliance controls:
  - Ensure all data used by AI respects local privacy and educational standards

## 7. Moderation & Incident Management

- Real-time moderation of:
  - Messaging and live sessions (with AI assistance)
  - Student or parent reports (flagged profiles or sessions)
- Incident tagging system:
  - Admin log + notification + resolution status
  - Internal escalation channels (e.g. platform ban, refund, coaching)

## 8. Control Panel for Customization

- Admin can:
  - Customize dashboard colors / logos / structure per organization
  - Define academic paths and levels per country or curriculum
  - Push new content templates to teachers (course plans, chapter lists, etc.)

- Create seasonal campaigns (e.g. “Back to School” bundles or promotions)
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## ◆ F. Security, Compliance & Infrastructure

### 1. Compliance with International Regulations

- Full alignment with:
  - **GDPR** (EU)
  - **COPPA** (US – for under-13 users)
  - **Digital Services Act** (DSA – EU content & platform liability rules)
  - **FERPA** (US – for education data privacy, optional)
- Data retention policies customizable per country

### 2. Authentication & Identity Security

- **Multi-factor authentication (MFA)** for all teacher, admin, and parent accounts
- **Google OAuth / Apple Sign-in / Email & password**
- Device/session tracking with logout-on-other-device option
- Optional **biometric login** on mobile apps (Face ID / Touch ID)

### 3. Encryption & Storage

- **TLS 1.3 encryption** for all data in transit
- **AES-256 encryption** for all stored user data and content
- **Encrypted backups** daily (stored in multiple regions)
- Option to **self-host** in-country for legal compliance

### 4. Audit Trails & Logging

- Every action is logged:
  - Profile edits, payments, session history
  - AI-generated content usage and overrides
- Logs are exportable, filterable, and time-stamped
- Real-time alert system for suspicious activity (fraud detection, policy violations)

### 5. Infrastructure & Scalability

- Full-stack **TypeScript** (Next.js + Node.js + PostgreSQL)
- Hosted on:
  - **Vercel** (frontend – globally distributed, auto-scaling)
  - **Render** or **Railway** (backend – secure, scalable API infrastructure)
- **CDN (Content Delivery Network)** for media and static content
  - Brotli compression for speed
  - Edge Workers for local responsiveness
- **PWA capabilities** (Progressive Web App) for offline access on mobile/tablet

### 6. AI Governance & Ethical Use

- Optional admin approval of:
  - AI-generated content before publication
  - AI feedback reports to students/parents
- Explainability layer: AI outputs come with origin tags (e.g. "based on C8 Algebra")
- Parental controls:
  - Activate/deactivate AI tools per student
  - Access to transcripts of AI interactions

### 7. Disaster Recovery & Business Continuity

- System health monitored 24/7
- Recovery Point Objective (RPO):  $\leq 24$  hours
- Recovery Time Objective (RTO):  $\leq 2$  hours
- Real-time backup sync + rollback versioning

## 8. ISO 27001 Preparation & Documentation

- System and documentation designed to meet ISO/IEC 27001 readiness:
    - Risk management policy
    - Access control documentation
    - Incident response procedures
    - Regular security audits
- 

# ◆ G. Educational Content & AI Tools

## 1. AI-Generated Learning Resources

- **Pre-session tools:**
  - AI-generated **pre-quizzes** based on course checklist (semi-structured) or full curriculum (structured)
  - Suggested **flashcards** to preview concepts
- **Post-session tools:**
  - **Post-quizzes** with adaptive difficulty
  - Auto-generated **summaries** of the session
  - Personalized **remediation activities** (short videos, corrections, concept refresh)

## 2. Shared Learning Library (Community + Core)

- Publicly accessible (or private) database of:
  - Lesson **plans**
  - PDF **worksheets**
  - **Video links** (curated YouTube/other)
  - **Corrected exercises** (per subject/chapter/level)
  - Past **exam simulations**
- Teachers can upload resources under Creative Commons or private status
- Admin-curated **core library** for each curriculum level (C7, Grade 8, etc.)

## 3. Templates & Course Fillers for Teachers

- Structured templates:
  - Weekly or unit-based **lesson sheets**
  - **Exercise banks** with solutions
  - Pre-made **quiz formats** (copy-paste into Moodle/OpenEDx or internal system)
- Partially pre-filled materials:
  - Example: “Math – C8 – Geometry Week 2” includes 1 exercise, 1 flashcard group, 1 quiz to complete

## 4. Personalized Content Adaptation by AI

- AI proposes different difficulty levels based on:
  - Student level
  - Quiz results
  - Time spent per concept

- AI explains answers in **text, visual, or audio** form
- Generates **progressive series** of exercises per concept (e.g., 3 difficulty tiers)

## 5. External Evaluation Tools

- AI simulates standardized exams:
  - Generates MCQs, written responses, logic-based questions
  - Scores and categorizes answers by topic/subtopic
- **Self-evaluation interface:** students/parents can schedule progress tests
- **Certification readiness checks:**
  - For example: DELF (language), SAT, Baccalaureate, etc.

## 6. Learning Analytics & Recommendations

- AI suggests:
  - What lesson to take next
  - What skill to revise
  - Which class group to join
- Integrated with dashboard metrics:
  - “Your weak spot this month: Algebra”
  - “Top 3 topics you’re improving in: Geometry, Fractions, Ratios”

## 7. AI for Teachers

- Lesson drafting aid (draft + suggest content based on curriculum)
- Homework auto-correction
- Feedback summary generator (per student, per week)
- Alert system: student not progressing / low engagement / needs remediation

## 8. AI for Students

- Voice assistant for revision: “Explain Pythagorean theorem”
- Conversational practice (for language learning)
- Test simulation coach: AI conducts oral or MCQ sessions with timing
- Note generator: builds structured revision notes after each lesson

## 9. Teacher Collaboration Tools

- Shared planning boards per curriculum path
- Resource sharing between teachers working on same subject/level
- Option for co-teaching / teacher assistant roles

# ◆ H. Monetization & Premium Services

## 1. Commission-Based Model (1-on-1 Tutoring)

- **Standard mode (Free-form lessons):**
  - Commission on each completed session (e.g., 20–30%)
  - Transparent fee structure for teachers
- **Structured & Semi-Structured modes:**
  - Slightly higher platform commission due to value-added services (AI tools, learning plans)
  - Possibility to introduce sliding-scale commissions based on teacher quality & retention

## 2. Subscription Plans (For Families / Learners)



- **Free Plan:**
  - Access to free content, community forums, limited dashboard
  - 1 free AI diagnostic quiz per month
- **Standard Plan (€9–19/mo):**
  - Full access to AI quizzes, learning dashboards
  - Priority booking for tutors
  - Monthly progress report to parents
- **Premium Plan (€29–49/mo):**
  - Full access to all AI tools (flashcards, simulations, summaries)
  - Unlimited evaluations
  - VIP customer support
  - Invitation to group classes or webinars

### 3. Add-on: “Learn with AI” Pack (Optional per Session)

- Students/parents can purchase **per-session AI assistance**:
  - Pre-quiz + post-quiz
  - AI-generated summary
  - Targeted revision flashcards
  - Cost: small per-session fee (€1–3)
- Sold as:
  - Individual add-on at checkout
  - Bundle packs (e.g., 10 sessions = €20)

### 4. Group Class Packages

- Pricing per **student seat** (ex: €6–12 per class per student)
- Or per **class unit** (e.g., €99 for 10 students = 10 classes)
- Custom pricing for:
  - Long-term group learning contracts (e.g., 12 weeks)
  - Intensive revision camps (pre-exam)
- Includes:
  - Collaborative dashboard
  - Group live sessions
  - Shared progress tracking

### 5. Certification & Test Preparation Services

- **Mock exam packages:**
  - Auto-generated by AI
  - Graded and analyzed per skillset
  - Pay-per-exam or included in Premium plans
- **Official certification assistance:**
  - Admin support for registering with recognized boards
  - Document preparation + pre-cert tests
  - Optional live prep classes (bundled or individual)

### 6. B2B Services for Organizations

- **Corporate Learning Plans:**
  - Custom training paths
  - Employee dashboards
  - Monthly performance reports to HR
  - Options: 1-on-1 or group format
- **Partner Licensing / White Label:**

- Full access to DESTAVIA LMS + branding
- For use in schools, universities, training centers
- **Revenue-Sharing Partnerships:**
  - Co-managed courses
  - % on sessions sold through institutional clients

## 7. Teacher Earnings & Incentives

- Transparent dashboard: sessions taught, AI usage, satisfaction score
- **Bonus system:**
  - High student retention
  - Excellent feedback
  - Contribution to shared content library
- **Referral rewards:**
  - Bring new students or teachers to the platform

## 8. Other Premium Services

- AI-powered monthly analytics report for parents
- Teacher performance report generation (for institutions)
- Paid marketing boosts (highlighting tutor profiles)
- Virtual classroom recording archive (with notes/transcripts)

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# ◆ I. Communication & Marketing

## 1. Multilingual Public Website (SEO-Optimized)

- **Landing pages** for each user group: students, parents, teachers, schools, companies.
- **SEO content** in multiple languages (FR, EN, AR, IT, ES) for organic traffic.
- **Geolocated content:** adapt offers and language to user region automatically.
- Core pages:
  - Home
  - How it works
  - Explore teachers
  - Pricing & Plans
  - Join as teacher
  - Enterprise
  - FAQs
  - Blog (with AI-generated articles)

## 2. Acquisition Funnels

- **Students/Parents:**
  - Lead magnet (free quiz, free session, diagnostic)
  - Clear pricing, benefits of AI-powered learning
  - Testimonials, success stories
- **Teachers:**
  - Simple onboarding flow (profile builder + guide)
  - Value proposition: autonomy + tools + visibility
- **B2B Clients (Schools & Companies):**

- Dedicated landing pages with case studies
- Request demo or call form
- White-label partnership section

### 3. Guided Onboarding Process

- **For students & parents:**
  - Welcome wizard: select grade, subject, goal
  - First diagnostic quiz
  - Suggested teachers + content packs
  - Dashboard walkthrough
- **For teachers:**
  - Profile quality checklist
  - Video tips for creating content
  - Certification upload and approval tracking
  - “First session” simulator tool
- **For companies:**
  - Setup wizard for team
  - Add employees, assign roles
  - Choose package or request custom plan

### 4. Community Engagement

- **Ambassador Program:**
  - Recruit top students, parents, and teachers as brand reps
  - Referral rewards
  - Early access to new features
- **Social Media Strategy:**
  - Short videos: testimonials, tutor highlights, micro-lessons
  - Contests: best AI score of the week, quiz challenges
  - Engagement: polls, Q&A sessions with experts
- **User-generated Content:**
  - Allow users to submit mini-lessons or reviews
  - Featured student wall

### 5. Educational Marketing Tools

- **Promo Packs for Schools:**
  - Printable leaflets
  - Slideshows for classroom presentation
  - Email sequences for parents
- **Video Content:**
  - Explainers (How to use DESTAVIA)
  - Case studies (success stories)
  - Tutorials (How to create your first quiz, etc.)
- **Webinars & Events:**
  - Monthly onboarding webinars
  - Teacher training sessions
  - Live Q&A with platform developers or AI experts

### 6. Tracking & Analytics

- Integration with analytics platforms (Google Analytics, Mixpanel)
- Email & SMS campaign performance tracking
- Conversion funnels + A/B testing of landing pages

- UTM tracking for campaigns

## 7. Help & Support Content

- Knowledge base in 5 languages
  - Video tutorials embedded in dashboard
  - Searchable FAQ with AI support assistant
  - Contextual help: guided tooltips and walkthroughs
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## ◆ J. B2B Strategy & Ecosystem

### 1. Corporate Training Packages

- DESTAVIA offers **tailored learning solutions** for companies:
  - **Individual mode** (1-on-1, Preply-style): targeted skill development per employee
  - **Group mode** (Moodle-style): internal virtual classrooms for teams
- Options include:
  - Language learning (English, French, Arabic, etc.)
  - Business skills (communication, management, IT tools)
  - Custom courses developed by companies or external tutors
- AI-generated:
  - Pre-training diagnostics per employee
  - Personalized content recommendations
  - Post-training performance report

### 2. Sponsored Group Classes (Schools / NGOs / Governments)

- **Centralized dashboards** for institutional partners:
  - Manage hundreds of learners
  - Assign courses and track attendance/progress
  - Monitor tutor performance and engagement
- **Whiteboard + live class** features for group sessions
- Optional integrations:
  - SCORM / Moodle LMS export
  - Student accounts synchronized with school systems
- Ideal for:
  - After-school support programs
  - State education campaigns
  - NGO literacy/skills initiatives

### 3. White-Label Platform Licensing

- DESTAVIA's infrastructure can be **licensed as a white-label** platform:
  - Custom domain and branding
  - Partner-specific configurations
  - Optional admin-level access to manage content and teachers
- Ideal for:
  - Tutoring companies seeking scalability
  - Ministries of Education seeking national-scale deployment
  - Educational startups needing an “off-the-shelf” EdTech base

### 4. Strategic Partnerships

- DESTAVIA will actively seek partnerships with:
  - **Universities:** for tutoring, test prep, and onboarding new students
  - **Educational publishers:** to integrate certified content into DESTAVIA templates
  - **Tech companies:** to co-develop AI learning modules or support integrations (e.g., Google for Education)

## 5. External Integrations & API Access

- **Open API** for integration with:
  - Google Classroom / Microsoft Teams
  - Moodle and other LMS systems
  - HR platforms for corporate training
- Sample use cases:
  - A university importing DESTAVIA progress reports into its student portal
  - A company embedding DESTAVIA quizzes into its onboarding platform
  - A local school using DESTAVIA's AI tools while keeping its internal content

## 6. Compliance & Customization

- B2B partners can request:
  - **Custom compliance** (local privacy laws, accessibility standards)
  - **Analytics dashboards** specific to their KPIs
  - **Admin hierarchies** (managers, departments, sub-teams)

## 7. Revenue Models (B2B)

- **Per-user license** (monthly or annual)
  - **Per-classroom pricing** (group packages)
  - **Custom SLAs** for white-label clients
  - **Revenue-sharing** for integrated content or branded programs
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